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(54) **NUCLEIC ACID LIGASE**

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(57) **ABSTRACT**

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§ 371 (c)(1),

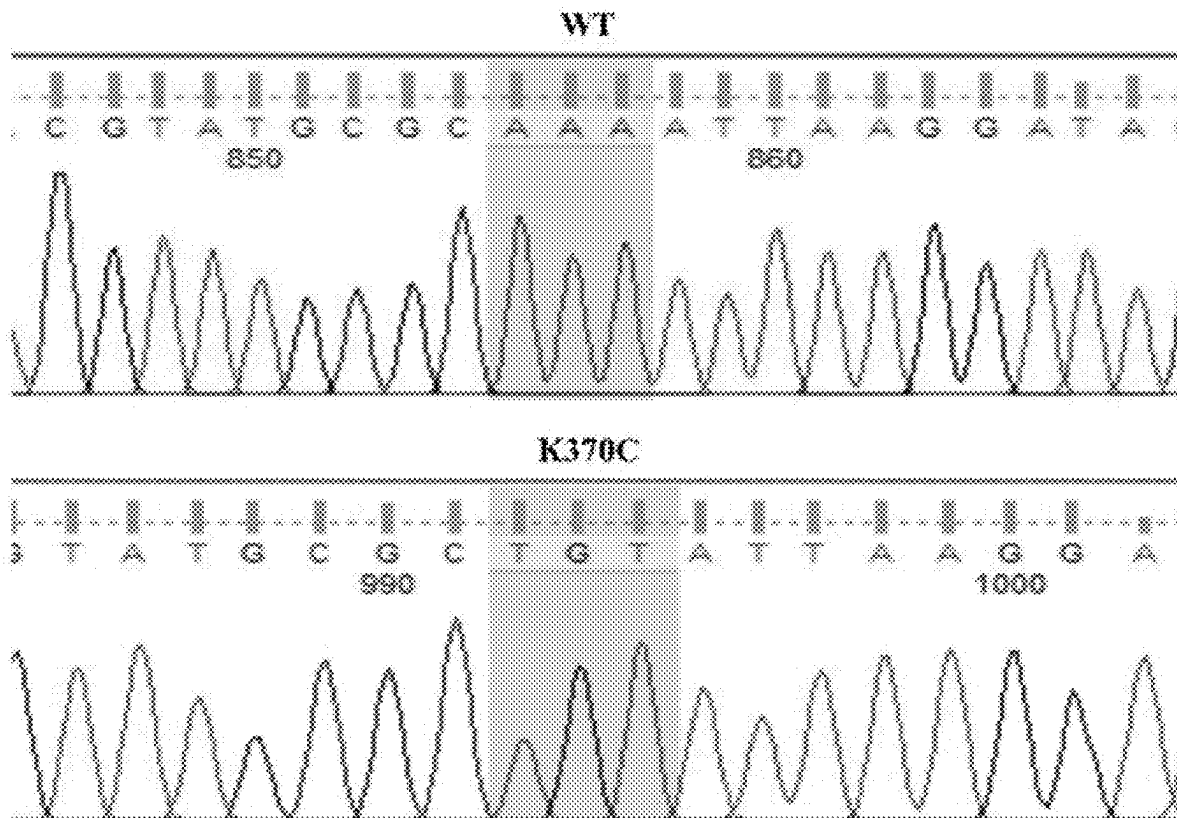
(2) Date: **Dec. 15, 2023**

Provided is a nucleic acid ligase, which comprises an amino acid sequence having a mutation at one or more positions selected from positions 79, 281, 370 and 372 compared to the amino acid sequence of Hyperligase of the prior art (SEQ ID NO: 1). Also provided are a nucleic acid molecule encoding the enzyme, a vector comprising the nucleic acid molecule, and a recombinant cell comprising the nucleic acid molecule or the vector. Also provided are a composition containing the enzyme and a use of the enzyme.

(30) **Foreign Application Priority Data**

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Specification includes a Sequence Listing.



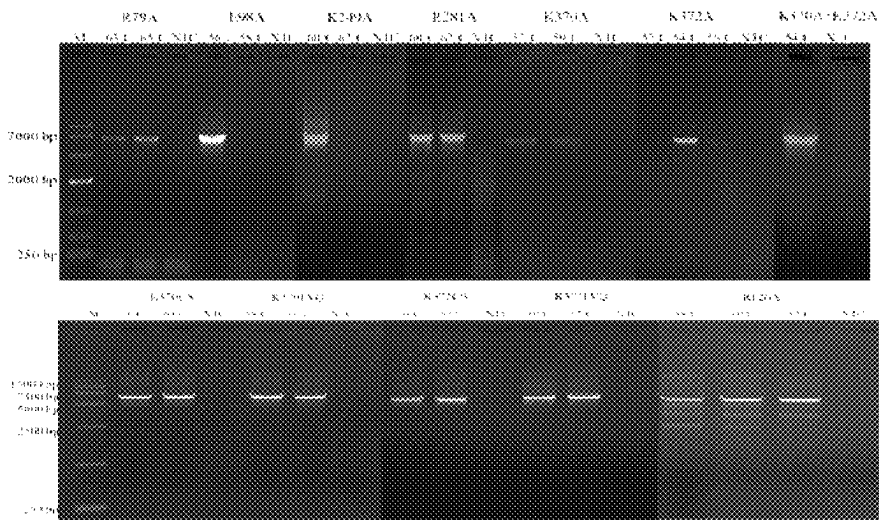


Fig.1

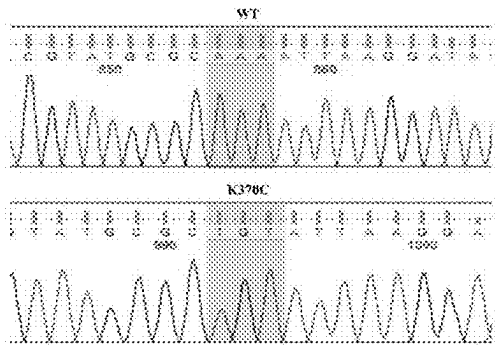


Fig.2

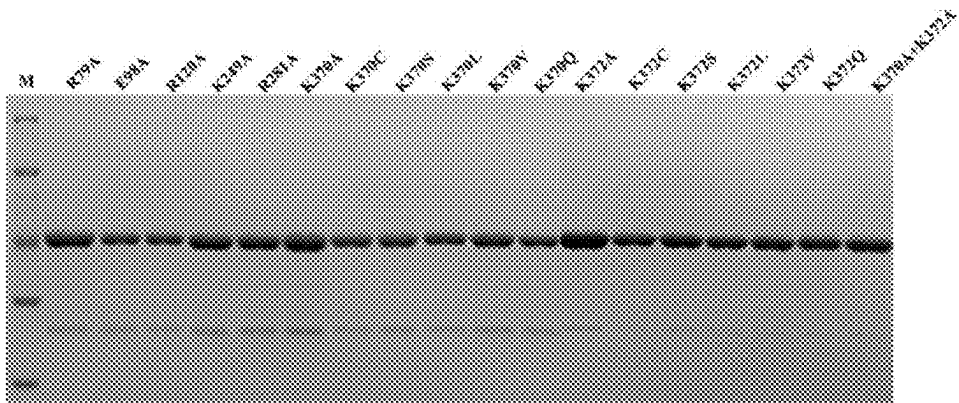


Fig.3

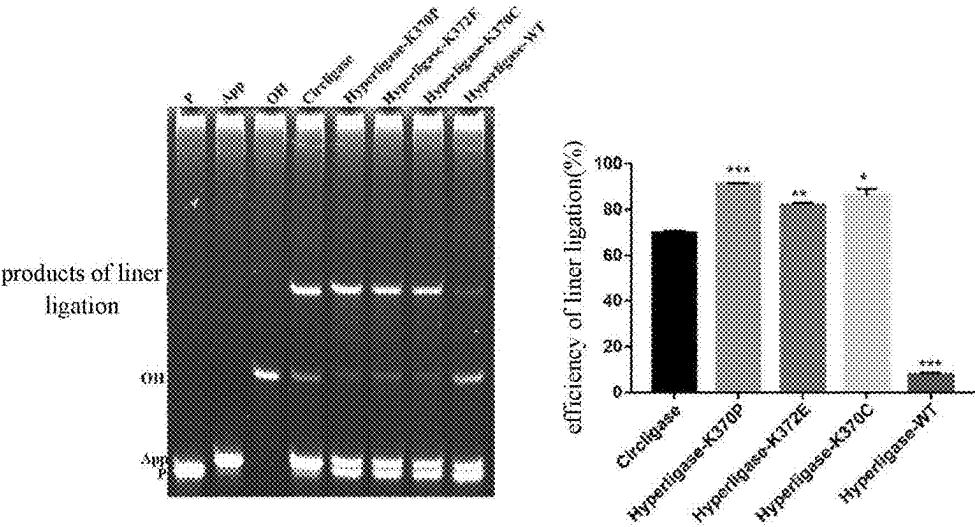


Fig.4

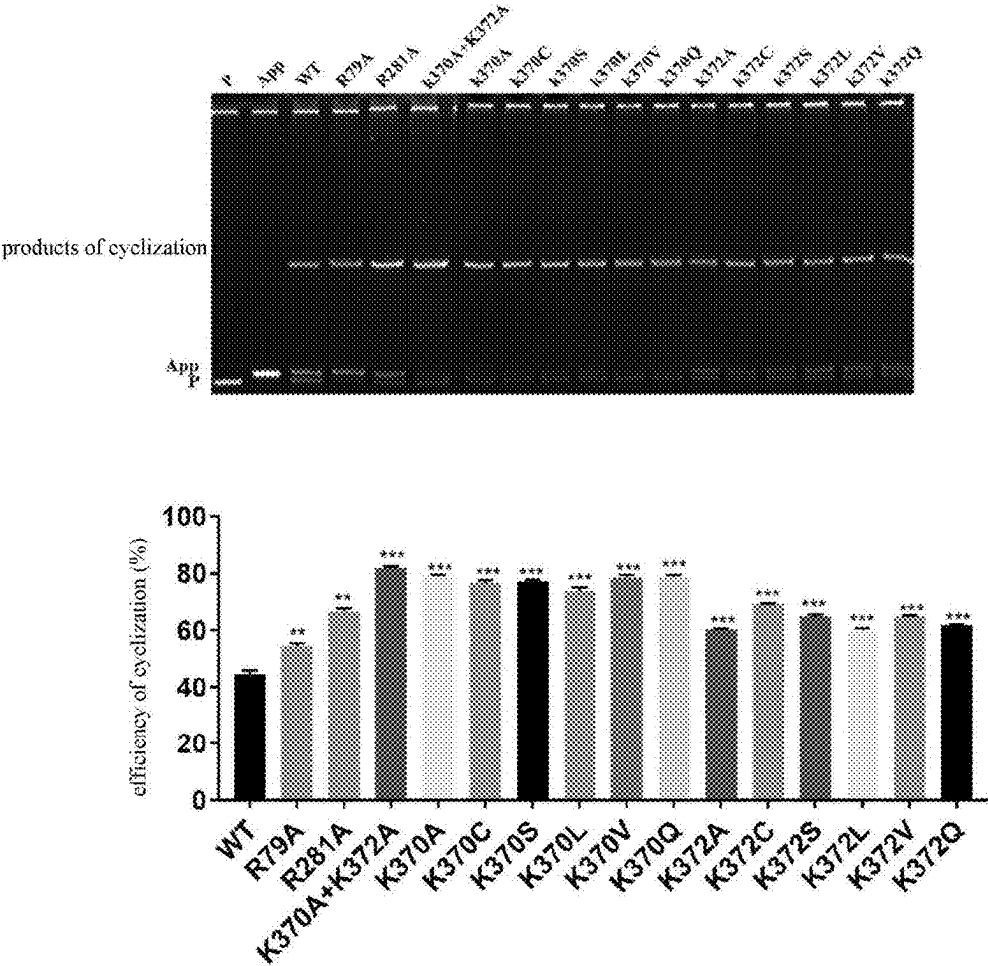


Fig.5

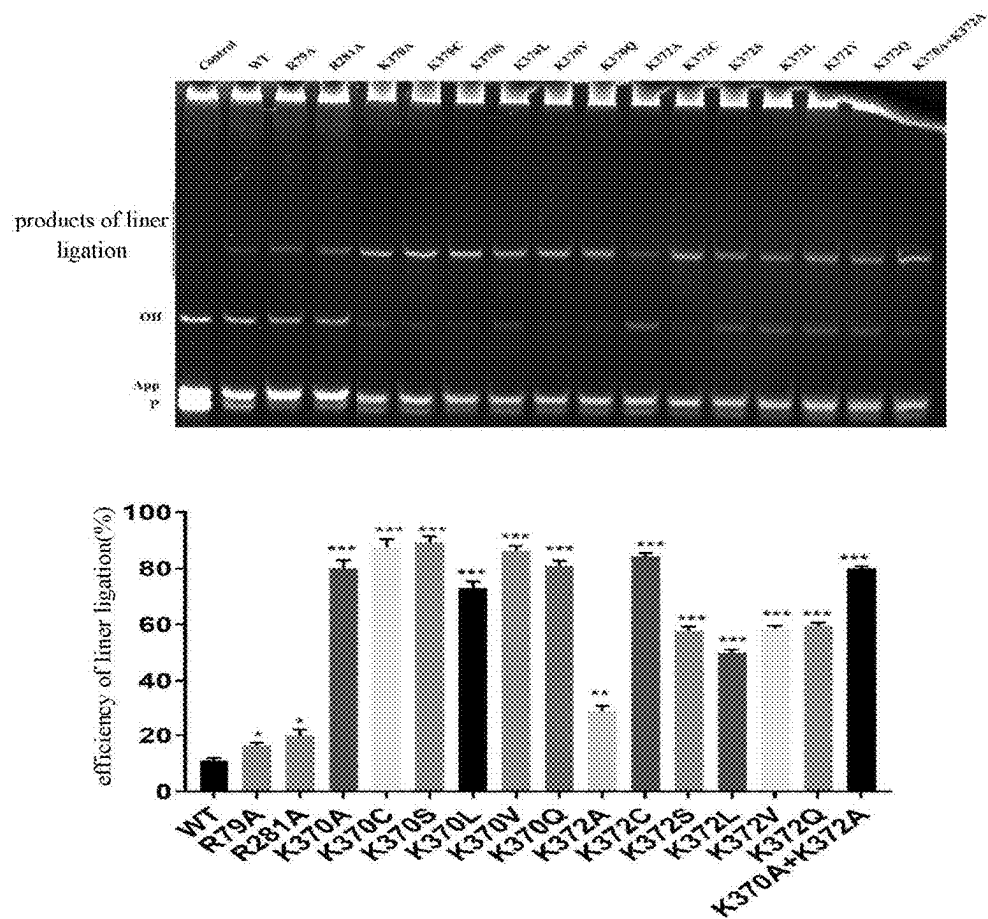


Fig.6

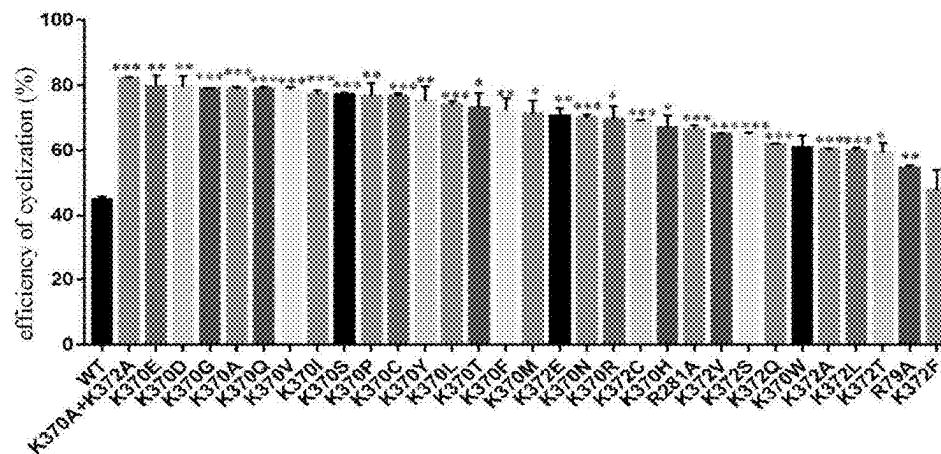


Fig.7

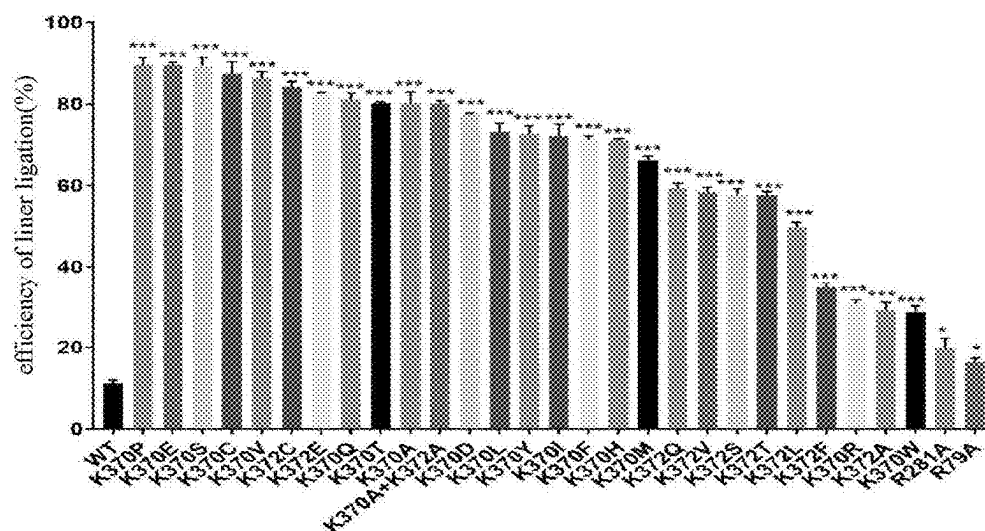


Fig.8

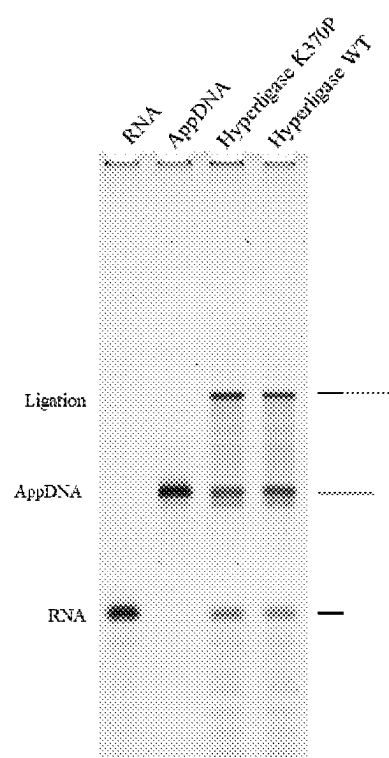


Fig.9

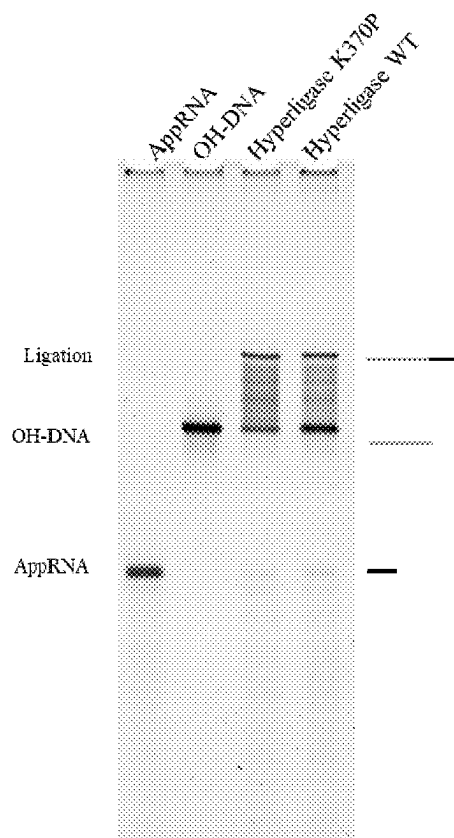


Fig.10

NUCLEIC ACID LIGASE

TECHNICAL FIELD

[0001] The present invention relates to the field of biotechnology. Specifically, it relates to new nucleic acid ligases obtained by mutating a nucleic acid ligase using current techniques. The invention also relates to a product comprising of said new enzymes.

BACKGROUND

[0002] Nucleic acid ligases are metal ion-dependent enzymes that catalyze the formation of a phosphodiester bond between the adjacent 3' end and 5' end of DNA or RNA molecules to achieve intramolecular cyclization or intermolecular linear ligation of a substrate sample. Nucleic acid ligases can be divided into DNA ligases and RNA ligases according to their catalytic substrates. Some ligases can catalyze the ligation reactions between DNA and RNA^[1, 2]. Depending on what species they are derived from, natural nucleic acid ligases have a variety of unique properties, such as substrate specificity, sequence preference, thermal stability, salt tolerance, pH change tolerance, etc.

[0003] It is generally believed that a complete nucleic acid ligation reaction is divided into three steps^[1]. In the first step, the ligase binds to the adenosine group of ATP or NAD⁺, transfers the adenosine group in ATP or NAD⁺ to a lysine residue in a conserved motif of the enzyme, releases a pyrophosphate, and forms an intermediate between the enzyme and the adenosine. This step is an equilibrium reaction and can proceed in both directions. In the second step, the intermediate between the enzyme and the adenosine transfers its adenosine group to the 5' phosphate end of a nucleic acid substrate to form a 5' adenylated intermediate. This step is also an equilibrium reaction and can proceed in both directions. In the third step, the lysine catalytic site of the enzyme binds to the 3' hydroxyl end of the nucleic acid, attacks the 5' adenylated end of the nucleic acid, catalyzes the formation of a phosphodiester bond between the two, and completes the ligation reaction. Common ligases generally use 5' phosphorylated nucleic acids as substrates to catalyze all three steps of the ligation reaction to complete the ligation reaction^[3, 4].

[0004] Ligases are important tools in molecular biology research and molecular diagnostic technology since they are the basis of common molecular biology methods such as molecular cloning technology, high-throughput sequencing library preparation, gene synthesis and molecular diagnosis^[5, 6]. According to the reaction temperature of the enzymes, ligases can be divided into two types: thermostable and non-thermostable. Currently, there are many ligases used for double-stranded DNA or RNA ligation reactions, but there are extremely limited choices for enzymes used for single-stranded DNA ligation reactions. There are even fewer thermostable ligases which can perform single-stranded ligation at higher temperatures (such as >65° C.).

[0005] Several thermostable single-stranded nucleic acid ligases have been reported. The thermostable TS2126 ligase (trade name: CircLigase) can directly catalyze the ligation reaction between 5' phosphate end and 3' hydroxyl end at 65° C.^[7], but the catalysis has end sequence preference, resulting in selective ligation of the substrates^[7], which leads to bias in the results. The genetically mutated Mth RNA thermostable ligase (trade name: Thermostable

5'AppDNA&RNA Ligase) can also catalyze the ligation reaction between single-stranded nucleic acids at higher temperatures, but its ligation efficiency is lower when used in single-stranded DNA ligation reactions^[8]. Taq DNA ligase, another enzyme that can catalyze the ligation reaction between single-stranded DNAs at higher temperatures, has higher ligation efficiency, but it requires the complementary strand as a template for guidance, and cannot be used alone in the ligation reaction of single-stranded nucleic acids.

[0006] Engineering and modification of enzymes mainly include genetic mutation, gene fusion, chemical modification, antibody modification, and nucleic acid aptamer modification, etc. Genetic engineering of some ligases can change the characteristics of the ligases such as improving the thermal stability or ligation efficiency, and the salt tolerance or acid tolerance. Mutation of the core lysine of the Mth RNA ligase allows it to directly use pre-adenylated single-stranded DNAs as substrates to perform ligation reactions to improve its ligation efficiency^[8]. Genetic mutation of the core lysine of the protein encoded by the Hbut_1550 gene from *Hyperthermus butylicus* confers the protein the ability to ligate single-stranded DNA/RNAs, so researchers engineered it into a heat-stable DNA/RNA Ligase (named Hyper-Thermostable Lysine-Mutant ssDNA/RNA Ligase, hereinafter referred to as HyperLigase, WO2017160788A3). HyperLigase ssDNA/RNA Ligase has high thermal stability and can catalyze the ligation reaction between the 5' pre-adenylated substrate and the 3' hydroxyl end in a wider temperature range (such as 37-95° C.). Despite this, the genetically engineered HyperLigase still suffer from insufficient ligation efficiency.

[0007] Therefore, there is still a need in the art for thermostable single-stranded nucleic acid ligases with higher ligation efficiency.

SUMMARY OF THE INVENTION

[0008] Based on a previous invention, by screening potential genetic mutation sites and carrying out genetic engineering, HyperLigase DNA/RNA ligases with new mutation sites were obtained in the present invention. Compared with the original HyperLigase DNA/RNA Ligase, these modified HyperLigase DNA/RNA ligases have higher enzymatic activities and improved catalytic activities.

[0009] Specifically, the present invention relates to the following aspects:

[0010] In one aspect, the present invention relates to a nucleic acid ligase, which comprises an amino acid sequence having mutations at one or more positions selected from positions 79, 281, 370 and 372 compared to the amino acid sequence of SEQ ID NO: 1. In one embodiment, the mutation at position 79 is the substitution of Arg with Ala, the mutation at position 281 is the substitution of Arg with Ala, the mutation at position 370 is the substitution of Lys with any natural amino acid except Lys, and the mutation at position 372 is the substitution of Lys with any natural amino acid except Lys. In one embodiment, the mutation at position 372 is the substitution of Lys with Glu, Cys, Val, Ser, Gln, Ala, Leu, Thr or Phe.

[0011] In one embodiment, the nucleic acid ligase of the invention comprises an amino acid sequence selected from SEQ ID NOs: 2-7.

[0012] In one embodiment, the nucleic acid ligase of the invention comprises the amino acid sequence of SEQ ID NO: 2 or 3, wherein Xaa is a non-polar amino acid,

preferably Ala. In one embodiment, the nucleic acid ligase of the invention comprises the amino acid sequence of SEQ ID NO: 4 or 5.

[0013] In one embodiment, the nucleic acid ligase of the invention comprises the amino acid sequence of SEQ ID NO: 6. In one embodiment, Xaa at positions 370 and 372 of SEQ ID NO: 6 is Ala.

[0014] In one embodiment, the nucleic acid ligase of the present invention comprises the amino acid sequence of SEQ ID NO: 7, wherein Xaa at positions 79 and 281 is a non-polar amino acid, preferably Ala. In one embodiment, Xaa at positions 370 and 372 of SEQ ID NO: 7 is a non-polar amino acid, preferably Ala.

[0015] In another aspect, the invention relates to a nucleic acid molecule encoding the nucleic acid ligases of the invention.

[0016] In another aspect, the invention relates to a vector comprising a nucleic acid molecule encoding the nucleic acid ligase of the invention.

[0017] In another aspect, the invention relates to recombinant cells into which the nucleic acid of the invention or the vector of the invention has been introduced.

[0018] In another aspect, the invention relates to a composition for ligating single-stranded DNAs and/or RNAs, which comprises a nucleic acid ligase of the invention.

[0019] In another aspect, the present invention relates to a kit for ligating single-stranded DNAs and/or RNAs, which comprises the nucleic acid ligase of the present invention.

[0020] In another aspect, the invention relates to use of a nucleic acid ligase of the invention in the preparation of a product for ligating single-stranded DNAs and/or RNAs.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] FIG. 1 shows the electrophoretic analysis of the mutant HyperLigase gene fragment amplified by inverse PCR (taking part of the results as an example).

[0022] FIG. 2 shows the capillary sequencing results for confirmation of wild-type and mutant HyperLigases (taking the K370C site as an example).

[0023] FIG. 3 shows the SDS-PAGE electrophoresis analysis of the purified mutant HyperLigase (taking part of the results as an example).

[0024] FIG. 4 shows the linear ligation reaction activity comparison among mutant HyperLigase, wild-type HyperLigase and CirLigase. Each data is the average of three independent replicate experiments; T test; *: $p < 0.05$; **: $p < 0.01$, ***: $p < 0.001$.

[0025] FIG. 5 shows the comparison of the single-stranded cyclization reaction activities of mutant HyperLigase and wild-type HyperLigase. Each data is the average of three independent replicate experiments; T test; *: $p < 0.05$; **: $p < 0.01$, ***: $p < 0.001$.

[0026] FIG. 6 shows the comparison of the linear ligation activities of mutant HyperLigase and wild-type HyperLigase. Each data is the average of three independent replicate experiments; T test; *: $p < 0.05$; **: $p < 0.01$, ***: $p < 0.001$.

[0027] FIG. 7 shows a summary of the cyclization efficiency of all mutant Hyperligases. The figure shows a summary of the cyclization efficiencies of all mutant Hyperligases ranked from high to low; each data is the average of three independent replicate experiments; T test; *: $p < 0.05$; **: $p < 0.01$, ***: $p < 0.001$.

[0028] FIG. 8 shows a summary of the linear ligation efficiency of all mutant Hyperligases. The figure shows a

summary of the linear ligation efficiencies of all mutant Hyperligases ranked from high to low; each data is the average of three independent replicate experiments; T test; *: $p < 0.05$; **: $p < 0.01$, ***: $p < 0.001$.

[0029] FIG. 9 compares the activities between the wild-type Hyperligase and the K370P mutant on the ligation between 3' hydroxyl end of RNA and the 5' adenylated end of DNA, showing that the K370P mutation enhances the activity of Hyperligase on the ligation between 3' hydroxyl end of RNA and the 5' adenylated end of DNA.

[0030] FIG. 10 compares the activities between the wild-type Hyperligase and the K370P mutant on the ligation between 3' hydroxyl end of DNA and the 5' adenylated end of RNA, showing that the K370P mutation enhances the activity of Hyperligase on the ligation between 3' hydroxyl end of DNA and the 5' adenylated end of RNA.

DETAILED DESCRIPTION OF THE INVENTION

[0031] The following examples are provided to show preferred embodiments. Those skilled in the art will recognize that the technologies disclosed in the following examples represent those discovered by the inventors to function well in the practice of the methods disclosed herein, and thus may be considered to constitute preferred modes of practice. In light of the present disclosure, however, those skilled in the art will recognize that many changes can be made in the specific embodiments disclosed without departing from the principles and scope of the methods disclosed herein and still obtain the same or similar results.

Example 1

Construction of a Protein Expression Plasmid with Mutant HyperLigase Using Site-Directed Mutagenesis

[0032] Starting from the HyperLigase of the prior art (SEQ ID NO: 1), the applicant conducted analysis and research and screened out potential mutation sites, namely Arg79, Lys249, Lys370 and Lys372 of HyperLigase.

[0033] Based on the screened mutation sites and the corresponding amino acids to be modified, the NEBasechanger site-directed mutation primer online design tool (<https://nebasechanger.neb.com>) was used to design back-to-back point mutation primers. After synthesizing the corresponding primers, the wild-type HyperLigase protein (SEQ ID NO: 1) expression plasmid was used as a template and a high-fidelity DNA polymerase was used to perform a PCR reaction. After the product was analyzed by electrophoresis to have the correct size, fragment purification was performed (FIG. 1). After purification, the DNA was terminally phosphorylated and circularized. The product was transformed into BL21 (DE3) competent cells and plated on a plate containing antibiotics. The colonies obtained by antibiotic screening were selected and sequenced to confirm that the sequence was correct (see FIG. 2). Mutant HyperLigase protein expression plasmids were obtained. The above procedures all used commercial reagents and were performed in accordance with the instructions of the kit. HyperLigase protein was expressed with a His tag located at the N-terminus or C-terminus of the target protein.

[0034] In addition to the above single mutations, two sites, K370 and K372, were selected for co-mutation to construct a double mutated K370A+K372A enzyme in which both sites were simultaneously mutated to alanine.

Example 2

Expression Induction and Purification of the Proteins

[0035] Positive colonies were picked from the plate and were inoculated into LB medium for recovery. After recovery, the culture was expanded by 10 to 50 times and then transferred to one Liter LB medium and cultured until the OD600 absorbance was 0.6 to 0.8, so that the bacteria reached the logarithmic growth phase. IPTG inducer was added to a final concentration of 0.1 to 1 mmol/L, and the culture was induced and cultured at 14 to 37° C. for 6 to 24 hours. The bacterial cells were collected and precipitated, and then were broken by ultrasonic treatment, and the broken products were purified. Purification was carried out using manual or instrumental methods, following the instructions for the equipment's. After elution, the protein concentration was measured and converted to molar concentration, and SDS-PAGE was used for purity identification (FIG. 3). The mutant HyperLigases were stored in 50% glycerol solution at -20° C. for long-term storage.

Example 3

Comparison of Activities Among Mutated HyperLigase Enzymes, the Original Hyperligase and CircLigase Enzymes

[0036] CircLigase is the most active single-stranded DNA ligase currently known on the market. In order to test the activities of HyperLigases before and after mutation, and CircLigase, an intermolecular linear ligation reaction was used to compare the activities of the enzymes.

[0037] The CircLigase linear ligation reaction system was as follows:

reagent	Final concentration	1× (μL)
10× CircLigase II Buffer	1×	2
50 mM MnCl ₂	2.5 mM MnCl ₂	1
5'-rApp/3'-blocked oligo	0.5 μM	1
3'-OH oligo	0.1 μM	1
PEG8000, 50% w/v	20%	8
CircLigase	50 U	0.5
H ₂ O	/	To 20 μL

[0038] The HyperLigases linear ligation reaction system was as follows:

reagent	Final concentration	1× (μL)
10× HyperLigase Buffer, pH = 7.5	70 mM Tris-HCl	1
20× MnCl ₂	10 mM MnCl ₂	0.5
5'-rApp/3'-blocked oligo	0.5 μM	1
3'-OH oligo	0.1 μM	1
PEG8000, 50% w/v	20%	8
Wild-type or mutant HyperLigase	0.5 μM	1
H ₂ O	/	To 20 μL

[0039] The HyperLigase reaction was carried out at 75° C., and the CircLigase reaction was carried out at 60° C. The reaction time was 6 hours for both. At the end of the reaction, Urea-PAGE electrophoresis and grayscale analysis were carried out to compare the difference in enzyme activities (FIG. 4).

[0040] The test results showed that the K370P mutation, K372E mutation, and K370C mutation all significantly

improved the catalytic activity of HyperLigase, and the activity of HyperLigase after mutation was significantly higher than that of CircLigase.

Example 4

Activity Test of Mutant HyperLigase Enzymes

(1) Circularization Efficiency Test

[0041] Using the synthesized pre-adenylated substrates, the intramolecular circularization reaction system was prepared according to the following table:

reagent	Final concentration	1× (μL)
10× HyperLigase Buffer, pH = 7.5	70 mM Tris-HCl	1
20× MnCl ₂	10 mM MnCl ₂	0.5
5'-rApp oligo	0.1 μM	1
Wild-type or mutant HyperLigase	0.1 μM	various
H ₂ O	/	To 10 μL

[0042] The reaction conditions were 60 to 75° C. for 6 hours, and maintained at 4° C.

[0043] Reaction product detection: The circularized products were identified by Urea-PAGE electrophoresis. The catalytic activities of the enzymes were characterized according to the change in the positions of the electrophoresis bands. The difference in the catalytic activities of the enzymes were compared according to the ratio of the grayscale values of the circularized product band to the substrate band as catalyzed by different mutant enzymes. The results are shown in FIG. 5.

[0044] The results of the single-stranded DNA circularization reaction showed that the R79, R281, K370, or K372 to alanine mutants significantly improved the activity, and the double mutant of K370 and K372 to alanine improved the catalytic activity. At the same time, for positions K370 and K372, mutation of lysine to other amino acids all improved the catalytic activity.

(2) Linear Ligation Efficiency Test

[0045] Using a synthetic substrate with adenylation at the 5' end and a blocking modification at the 3' end, and a substrate with a hydroxyl group at the 3'end, a linear ligation reaction system was prepared according to the following table:

reagent	Final concentration	1× (μL)
10× HyperLigase Buffer, pH = 7.5	70 mM Tris-HCl	2
20× MnCl ₂	10 mM MnCl ₂	1
5'-rApp/3'-blocked oligo	0.5 μM	1
3'-OH oligo	0.1 μM	1
PEG8000, 50% w/v	20%	8
Wild-type or mutant HyperLigase	0.5 μM	various
H ₂ O	/	To 20 μL

[0046] The reaction conditions were 60 to 75° C. for 6 hours, and then maintained at 4° C.

[0047] Reaction product detection: The linear ligation product was analyzed by Urea-PAGE electrophoresis. The catalytic activities of the enzymes were characterized according to the change in the positions of the electrophoresis band. The difference in the catalytic activities of the

enzymes were compared according to the ratio of the grayscale values of the liner ligation product band to the 3'-OH end substrate band as catalyzed different mutant enzymes. The results are shown in FIG. 6.

[0048] The linear ligation reaction test results showed that the enzyme activity was significantly improved after any of the four amino acids of R79, R281, K370, and K372 was mutated to alanine, or both the K370 and K372 were mutated to alanine. At the same time, for K370 and K372 positions, mutation of lysine to other amino acids improved the catalytic activity of the enzyme.

[0049] Furthermore, when the K370 and K372 sites were mutated to other amino acids, the results showed that the activity was significantly improved compared to wild-type HyperLigase after mutating these two sites to any other amino acids (FIGS. 5 and 6).

[0050] For the activity test of the above mutation sites, the linear ligation reaction results were consistent with the single-stranded DNA circularization reaction results.

[0051] In summary, the mutations for any of the R79, R281, K370, and K372 residues or the combined mutations of K370 and K372 significantly improve the activity of the enzyme. Additionally, for the K370 and K372 positions, mutation of lysine to other amino acids improved the catalytic activity of the enzyme (see FIGS. 7 and 8).

Example 5

Test of the Activity of K370P Mutant Ligase in Ligating the 3'-Hydroxyl End of RNA and the 5' Adenylated End of DNA

[0052] A 21-base RNA sequence was selected as the acceptor, and a 35-base 5'-end pre-adenylated and 3'-end blocked DNA was selected as the donor. It was expected that the 3' hydroxyl end of the RNA was ligated to the 5' adenylated end of the DNA. The reaction system was as follows:

reagent	Initial concentration	Final concentration	1× (μL)
Tris-HCl, pH = 8	700 mM	70 mM	2
5'-rApp-CL78_cp5_UMI8	5 μM	0.25 μM	1
5P-21mer-RNA	1 μM	0.25 μM	5
PEG8000	50% w/v	20%	8
H ₂ O	/	/	1.74
Hyperligase K370P or WT	3.98 μM	0.25 μM	1.26
MnCl ₂	100 mM	5 mM	1
total		/	20

[0053] Reaction conditions: 65° C., 30 min/1 h; 4° C., hold. The reaction products were identified by Urea-PAGE electrophoresis.

[0054] The results are shown in FIG. 9, which shows that the K370P mutation enhanced the activity of Hyperligase on the ligation between the 3' hydroxyl end of RNA and the 5' adenylated end of DNA. It can be seen from the figure that the grayscale of the band of the K370P ligation product increased, and the substrate band of AppDNA was weaker than the wild-type, thus indicating the ligation activity of the enzyme was improved.

Example 7

Test of the Activity of K370P Mutant Ligase in Ligating the 3'-Hydroxyl End of DNA and the 5' Adenylated End of RNA

[0055] A 28-base RNA sequence was selected as the donor, and a 55-base single-stranded DNA with a hydroxyl group at the 3' end was selected as the acceptor. It was expected that the 3' hydroxyl end of the DNA was ligated to the 5' adenylated end of the RNA. The reaction system was as follows:

reagent	Initial concentration	Final concentration	1× (μL)
H ₂ O		/	5.9
Tris-HCl, pH = 8	700 mM	70 mM	2
5App3NH2-28mer-RNA	3.73	0.25	1.34
OH-5N3N-Cir-Con	10 μM	0.5 μM	0.5
PEG8000	50% w/v	20%	8
Hyperligase K370P or WT	3.98 μM	0.25 μM	1.26
MnCl ₂	100 mM	5 mM	1
total		/	20

[0056] Reaction conditions: 65° C., 30 min; 16° C., hold.

[0057] The results are shown in FIG. 10, which shows that the K370P mutation enhanced the activity of Hyperligase on the ligation between the 3' hydroxyl end of DNA and the 5' adenylated end of RNA. It can be clearly seen that the amount of OH-DNA substrate of K370P was reduced, indicating that more substrates were used for ligation. There was degradation in the ligation product, with stronger degradation bands derived from K370P ligase than from WT HyperLigase, also confirming better ligation efficiency of the former.

REFERENCES

[0058] [1] VIOLLET S, FUCHS R T, MUNAFO D B, et al. T4 RNA ligase 2 truncated active site mutants: improved tools for RNA analysis [J]. BMC Biotechnol, 2011, 11 (72).

[0059] [2] LOHMAN G J, TABOR S, NICHOLS N M. DNA ligases [J]. Curr Protoc Mol Biol, 2011, Chapter 3 (Unit3 14).

[0060] [3] JOHNSON A, O'DONNELL M. DNA ligase: Getting a grip to seal the deal [J]. Current Biology, 2005, 15 (3): R90-R2.

[0061] [4] CHEREPANOV A V, DE VRIES S. Dynamic mechanism of nick recognition by DNA ligase [J]. European Journal of Biochemistry, 2002, 269 (24): 5993-9.

[0062] [5] TUMBALE P P, JURKIW T J, SCHELLENBERG M J, et al. Two-tiered enforcement of high-fidelity DNA ligation [J]. Nature Communications, 2019, 10 (1): 5431.

[0063] [6] FOY CA, PARKES H C. Emerging Homogeneous DNA-based Technologies in the Clinical Laboratory [J]. Clinical Chemistry, 2001, 47 (6): 990-1000.

[0064] [7] BLONDAL T, THORISDOTTIR A, UNN-STEINSDOTTIR U, et al. Isolation and characterization of a thermostable RNA ligase 1 from a *Thermus scotoductus* bacteriophage TS2126 with good single-stranded DNA ligation properties [J]. Nucleic Acids Res, 2005, 33 (1): 135-42.

[0065] [8] ZHELKOVSKY A M, MCREYNOLDS L A. Structure-function analysis of *Methanobacterium thermoautotrophicum* RNA ligase-engineering a thermostable ATP independent enzyme [J]. BMC Molecular Biology, 2012, 13 (1): 10.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 7

<210> SEQ ID NO 1

<211> LENGTH: 390

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: HyperLigase

<400> SEQUENCE: 1

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Gly Thr Ile Arg Phe Arg Gly Tyr Arg Asp Val Val Tyr Ala Leu Leu
          35           40           45

Arg Arg Glu Ile Ala Gly His Pro Glu Gly Thr Val Val Leu Leu Glu
          50           55           60

Arg Asn Gly Ala Arg Leu Val Pro Gly Tyr Pro Pro Ile Gln Arg Met
65           70           75           80

Val Leu Pro Thr Ile Ala Leu Pro Arg His Phe Ile Asp Lys Val Val
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Leu Leu Gly Val Ile Asp Lys Asn Asp Val Glu Met Phe Lys Gln Ile
          210          215          220

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Pro Glu Tyr Arg Val Pro Pro Leu Lys Tyr Thr Thr Ser Ala Val Asn
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Glu Leu Glu Glu Phe Thr Ser Tyr Met Ala Glu Leu Gly Val Asp Ile
340 345 350

Val Val Val Ser Thr Ser Arg Glu Asp Glu Gly Leu Arg Ala Arg Met
355 360 365

Arg Lys Ile Lys Asp Thr Trp Ile Gln Leu Arg Lys Val Leu Asp Thr
370 375 380

Gly Leu Ser Pro Val Asp
385 390

<210> SEQ ID NO 2

<211> LENGTH: 390

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: R79 mutant of Hyperligase

<220> FEATURE:

<221> NAME/KEY: misc_feature

<222> LOCATION: (79)..(79)

<223> OTHER INFORMATION: Xaa can be any amino acid except Arg

<400> SEQUENCE: 2

Met Thr Trp Ile Lys Asn Pro Glu Pro Trp Met Val Asn Leu Val Ala
1 5 10 15

Glu Lys Leu Gly Leu Asp Val Glu Arg Val Glu Thr Leu Ala Arg His
20 25 30

Gly Thr Ile Arg Phe Arg Gly Tyr Arg Asp Val Val Tyr Ala Leu Leu
35 40 45

Arg Arg Glu Ile Ala Gly His Pro Glu Gly Thr Val Val Leu Leu Glu
50 55 60

Arg Asn Gly Ala Arg Leu Val Pro Gly Tyr Pro Pro Ile Gln Xaa Met
65 70 75 80

Val Leu Pro Thr Ile Ala Leu Pro Arg His Phe Ile Asp Lys Val Val
85 90 95

Val Glu Glu Ala Met Asn Gly Tyr Asn Val Arg Leu Val Met Phe His
100 105 110

Arg Lys Leu Leu Ala Val Thr Arg Gly Gly Phe Ile Cys Pro Tyr Thr
115 120 125

Thr Ala Arg Leu Glu Arg Leu Ile Gly Gly Arg Val Arg Glu Leu Phe
130 135 140

Arg Glu Ile Asp Pro Glu Thr Tyr Thr Ile Ala Gly Glu Val Val Gly
145 150 155 160

Leu Glu Asn Pro Tyr Thr Arg Tyr Phe Tyr Pro Glu Ala Pro Arg Phe
165 170 175

Asp Tyr Phe Val Phe Asp Leu Phe His Glu Leu Lys Pro Leu Pro Pro
180 185 190

Leu Glu Arg Asn Glu Leu Leu Glu Lys Tyr Gly Leu Lys His Val Arg
195 200 205

Leu Leu Gly Val Ile Asp Lys Asn Asp Val Glu Met Phe Lys Gln Ile
210 215 220

Val Ala Glu Leu Asp Arg Glu Gly Arg Glu Gly Val Val Ala Lys Asp

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225	230	235	240
Pro Glu Tyr Arg Val Pro Pro Leu Lys Tyr Thr Thr Ser Ala Val Asn	245	250	255
Ile Gly Asp Val Arg Tyr Gly Met Arg Phe Phe Met Glu Glu Gly Arg	260	265	270
Ser Phe Leu Phe Ser Arg Leu Leu Arg Glu Leu Phe Arg Ala Tyr Glu	275	280	285
Glu Gly Phe Gly Asp Ala Gln Leu Glu Lys Leu Ala Leu Glu Phe Gly	290	295	300
Arg Ala Ala Thr Glu Pro Ala Leu Glu Ser Ile Arg Lys Val Ala Met	305	310	315
Gly Asp Met Leu Tyr Glu Glu Phe Glu Leu Val Phe Ala Asp Glu Val	325	330	335
Glu Leu Glu Glu Phe Thr Ser Tyr Met Ala Glu Leu Gly Val Asp Ile	340	345	350
Val Val Val Ser Thr Ser Arg Glu Asp Glu Gly Leu Arg Ala Arg Met	355	360	365
Arg Lys Ile Lys Asp Thr Trp Ile Gln Leu Arg Lys Val Leu Asp Thr	370	375	380
Gly Leu Ser Pro Val Asp	385	390	

<210> SEQ ID NO 3
 <211> LENGTH: 390
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: R281 mutant of HyperLigase
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (281)..(281)
 <223> OTHER INFORMATION: Xaa can be any amino acid except Arg

<400> SEQUENCE: 3

Met Thr Trp Ile Lys Asn Pro Glu Pro Trp Met Val Asn Leu Val Ala	1	5	10	15
Glu Lys Leu Gly Leu Asp Val Glu Arg Val Glu Thr Leu Ala Arg His	20	25	30	
Gly Thr Ile Arg Phe Arg Gly Tyr Arg Asp Val Val Tyr Ala Leu Leu	35	40	45	
Arg Arg Glu Ile Ala Gly His Pro Glu Gly Thr Val Val Leu Leu Glu	50	55	60	
Arg Asn Gly Ala Arg Leu Val Pro Gly Tyr Pro Pro Ile Gln Arg Met	65	70	75	80
Val Leu Pro Thr Ile Ala Leu Pro Arg His Phe Ile Asp Lys Val Val	85	90	95	
Val Glu Glu Ala Met Asn Gly Tyr Asn Val Arg Leu Val Met Phe His	100	105	110	
Arg Lys Leu Leu Ala Val Thr Arg Gly Gly Phe Ile Cys Pro Tyr Thr	115	120	125	
Thr Ala Arg Leu Glu Arg Leu Ile Gly Gly Arg Val Arg Glu Leu Phe	130	135	140	
Arg Glu Ile Asp Pro Glu Thr Tyr Thr Ile Ala Gly Glu Val Val Gly	145	150	155	160

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Leu	Glu	Asn	Pro	Tyr	Thr	Arg	Tyr	Phe	Tyr	Pro	Glu	Ala	Pro	Arg	Phe
				165					170					175	
Asp	Tyr	Phe	Val	Phe	Asp	Leu	Phe	His	Glu	Leu	Lys	Pro	Leu	Pro	Pro
			180					185					190		
Leu	Glu	Arg	Asn	Glu	Leu	Leu	Glu	Lys	Tyr	Gly	Leu	Lys	His	Val	Arg
		195					200					205			
Leu	Leu	Gly	Val	Ile	Asp	Lys	Asn	Asp	Val	Glu	Met	Phe	Lys	Gln	Ile
	210					215					220				
Val	Ala	Glu	Leu	Asp	Arg	Glu	Gly	Arg	Glu	Gly	Val	Val	Ala	Lys	Asp
225					230					235					240
Pro	Glu	Tyr	Arg	Val	Pro	Pro	Leu	Lys	Tyr	Thr	Thr	Ser	Ala	Val	Asn
				245					250					255	
Ile	Gly	Asp	Val	Arg	Tyr	Gly	Met	Arg	Phe	Phe	Met	Glu	Glu	Gly	Arg
		260					265						270		
Ser	Phe	Leu	Phe	Ser	Arg	Leu	Leu	Xaa	Glu	Leu	Phe	Arg	Ala	Tyr	Glu
		275					280					285			
Glu	Gly	Phe	Gly	Asp	Ala	Gln	Leu	Glu	Lys	Leu	Ala	Leu	Glu	Phe	Gly
	290					295					300				
Arg	Ala	Ala	Thr	Glu	Pro	Ala	Leu	Glu	Ser	Ile	Arg	Lys	Val	Ala	Met
305					310					315					320
Gly	Asp	Met	Leu	Tyr	Glu	Glu	Phe	Glu	Leu	Val	Phe	Ala	Asp	Glu	Val
				325					330					335	
Glu	Leu	Glu	Glu	Phe	Thr	Ser	Tyr	Met	Ala	Glu	Leu	Gly	Val	Asp	Ile
			340					345					350		
Val	Val	Val	Ser	Thr	Ser	Arg	Glu	Asp	Glu	Gly	Leu	Arg	Ala	Arg	Met
		355					360					365			
Arg	Lys	Ile	Lys	Asp	Thr	Trp	Ile	Gln	Leu	Arg	Lys	Val	Leu	Asp	Thr
	370					375					380				
Gly	Leu	Ser	Pro	Val	Asp										
385					390										

<210> SEQ ID NO 4
 <211> LENGTH: 390
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: K370 mutant of Hyperligase
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (370)..(370)
 <223> OTHER INFORMATION: Xaa can be any amino acid except Lys
 <400> SEQUENCE: 4

Met	Thr	Trp	Ile	Lys	Asn	Pro	Glu	Pro	Trp	Met	Val	Asn	Leu	Val	Ala
1				5					10					15	
Glu	Lys	Leu	Gly	Leu	Asp	Val	Glu	Arg	Val	Glu	Thr	Leu	Ala	Arg	His
		20						25					30		
Gly	Thr	Ile	Arg	Phe	Arg	Gly	Tyr	Arg	Asp	Val	Val	Tyr	Ala	Leu	Leu
	35					40						45			
Arg	Arg	Glu	Ile	Ala	Gly	His	Pro	Glu	Gly	Thr	Val	Val	Leu	Leu	Glu
	50					55					60				
Arg	Asn	Gly	Ala	Arg	Leu	Val	Pro	Gly	Tyr	Pro	Pro	Ile	Gln	Arg	Met
65					70				75					80	
Val	Leu	Pro	Thr	Ile	Ala	Leu	Pro	Arg	His	Phe	Ile	Asp	Lys	Val	Val
			85						90					95	

-continued

Val Glu Glu Ala Met Asn Gly Tyr Asn Val Arg Leu Val Met Phe His
 100 105 110
 Arg Lys Leu Leu Ala Val Thr Arg Gly Gly Phe Ile Cys Pro Tyr Thr
 115 120 125
 Thr Ala Arg Leu Glu Arg Leu Ile Gly Gly Arg Val Arg Glu Leu Phe
 130 135 140
 Arg Glu Ile Asp Pro Glu Thr Tyr Thr Ile Ala Gly Glu Val Val Gly
 145 150 155 160
 Leu Glu Asn Pro Tyr Thr Arg Tyr Phe Tyr Pro Glu Ala Pro Arg Phe
 165 170 175
 Asp Tyr Phe Val Phe Asp Leu Phe His Glu Leu Lys Pro Leu Pro Pro
 180 185 190
 Leu Glu Arg Asn Glu Leu Leu Glu Lys Tyr Gly Leu Lys His Val Arg
 195 200 205
 Leu Leu Gly Val Ile Asp Lys Asn Asp Val Glu Met Phe Lys Gln Ile
 210 215 220
 Val Ala Glu Leu Asp Arg Glu Gly Arg Glu Gly Val Val Ala Lys Asp
 225 230 235 240
 Pro Glu Tyr Arg Val Pro Pro Leu Lys Tyr Thr Thr Ser Ala Val Asn
 245 250 255
 Ile Gly Asp Val Arg Tyr Gly Met Arg Phe Phe Met Glu Glu Gly Arg
 260 265 270
 Ser Phe Leu Phe Ser Arg Leu Leu Arg Glu Leu Phe Arg Ala Tyr Glu
 275 280 285
 Glu Gly Phe Gly Asp Ala Gln Leu Glu Lys Leu Ala Leu Glu Phe Gly
 290 295 300
 Arg Ala Ala Thr Glu Pro Ala Leu Glu Ser Ile Arg Lys Val Ala Met
 305 310 315 320
 Gly Asp Met Leu Tyr Glu Glu Phe Glu Leu Val Phe Ala Asp Glu Val
 325 330 335
 Glu Leu Glu Glu Phe Thr Ser Tyr Met Ala Glu Leu Gly Val Asp Ile
 340 345 350
 Val Val Val Ser Thr Ser Arg Glu Asp Glu Gly Leu Arg Ala Arg Met
 355 360 365
 Arg Xaa Ile Lys Asp Thr Trp Ile Gln Leu Arg Lys Val Leu Asp Thr
 370 375 380
 Gly Leu Ser Pro Val Asp
 385 390

<210> SEQ ID NO 5
 <211> LENGTH: 390
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: K372 mutant of HyperLigase
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (372)..(372)
 <223> OTHER INFORMATION: Xaa can be any amino acid except Lys
 <400> SEQUENCE: 5

Met Thr Trp Ile Lys Asn Pro Glu Pro Trp Met Val Asn Leu Val Ala
 1 5 10 15
 Glu Lys Leu Gly Leu Asp Val Glu Arg Val Glu Thr Leu Ala Arg His

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20					25					30					
Gly	Thr	Ile	Arg	Phe	Arg	Gly	Tyr	Arg	Asp	Val	Val	Tyr	Ala	Leu	Leu
	35					40					45				
Arg	Arg	Glu	Ile	Ala	Gly	His	Pro	Glu	Gly	Thr	Val	Val	Leu	Leu	Glu
	50				55						60				
Arg	Asn	Gly	Ala	Arg	Leu	Val	Pro	Gly	Tyr	Pro	Pro	Ile	Gln	Arg	Met
	65				70				75						80
Val	Leu	Pro	Thr	Ile	Ala	Leu	Pro	Arg	His	Phe	Ile	Asp	Lys	Val	Val
			85					90					95		
Val	Glu	Glu	Ala	Met	Asn	Gly	Tyr	Asn	Val	Arg	Leu	Val	Met	Phe	His
			100					105					110		
Arg	Lys	Leu	Leu	Ala	Val	Thr	Arg	Gly	Gly	Phe	Ile	Cys	Pro	Tyr	Thr
		115					120					125			
Thr	Ala	Arg	Leu	Glu	Arg	Leu	Ile	Gly	Gly	Arg	Val	Arg	Glu	Leu	Phe
		130				135					140				
Arg	Glu	Ile	Asp	Pro	Glu	Thr	Tyr	Thr	Ile	Ala	Gly	Glu	Val	Val	Gly
	145				150					155					160
Leu	Glu	Asn	Pro	Tyr	Thr	Arg	Tyr	Phe	Tyr	Pro	Glu	Ala	Pro	Arg	Phe
			165					170						175	
Asp	Tyr	Phe	Val	Phe	Asp	Leu	Phe	His	Glu	Leu	Lys	Pro	Leu	Pro	Pro
		180						185					190		
Leu	Glu	Arg	Asn	Glu	Leu	Leu	Glu	Lys	Tyr	Gly	Leu	Lys	His	Val	Arg
		195					200					205			
Leu	Leu	Gly	Val	Ile	Asp	Lys	Asn	Asp	Val	Glu	Met	Phe	Lys	Gln	Ile
		210				215					220				
Val	Ala	Glu	Leu	Asp	Arg	Glu	Gly	Arg	Glu	Gly	Val	Val	Ala	Lys	Asp
	225				230					235					240
Pro	Glu	Tyr	Arg	Val	Pro	Pro	Leu	Lys	Tyr	Thr	Thr	Ser	Ala	Val	Asn
			245						250					255	
Ile	Gly	Asp	Val	Arg	Tyr	Gly	Met	Arg	Phe	Phe	Met	Glu	Glu	Gly	Arg
		260					265						270		
Ser	Phe	Leu	Phe	Ser	Arg	Leu	Leu	Arg	Glu	Leu	Phe	Arg	Ala	Tyr	Glu
		275					280					285			
Glu	Gly	Phe	Gly	Asp	Ala	Gln	Leu	Glu	Lys	Leu	Ala	Leu	Glu	Phe	Gly
		290				295					300				
Arg	Ala	Ala	Thr	Glu	Pro	Ala	Leu	Glu	Ser	Ile	Arg	Lys	Val	Ala	Met
	305				310					315					320
Gly	Asp	Met	Leu	Tyr	Glu	Glu	Phe	Glu	Leu	Val	Phe	Ala	Asp	Glu	Val
			325					330						335	
Glu	Leu	Glu	Glu	Phe	Thr	Ser	Tyr	Met	Ala	Glu	Leu	Gly	Val	Asp	Ile
		340					345						350		
Val	Val	Val	Ser	Thr	Ser	Arg	Glu	Asp	Glu	Gly	Leu	Arg	Ala	Arg	Met
		355					360					365			
Arg	Lys	Ile	Xaa	Asp	Thr	Trp	Ile	Gln	Leu	Arg	Lys	Val	Leu	Asp	Thr
		370				375					380				
Gly	Leu	Ser	Pro	Val	Asp										
	385				390										

<210> SEQ ID NO 6

<211> LENGTH: 390

<212> TYPE: PRT

<213> ORGANISM: artificial sequence

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<220> FEATURE:
<223> OTHER INFORMATION: combined K370 and K372 mutant of HyperLigase
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (370)..(370)
<223> OTHER INFORMATION: Xaa can be any amino acid except Lys
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (372)..(372)
<223> OTHER INFORMATION: Xaa can be any amino acid except Lys

<400> SEQUENCE: 6

Met Thr Trp Ile Lys Asn Pro Glu Pro Trp Met Val Asn Leu Val Ala
1          5          10          15

Glu Lys Leu Gly Leu Asp Val Glu Arg Val Glu Thr Leu Ala Arg His
20          25          30

Gly Thr Ile Arg Phe Arg Gly Tyr Arg Asp Val Val Tyr Ala Leu Leu
35          40          45

Arg Arg Glu Ile Ala Gly His Pro Glu Gly Thr Val Val Leu Leu Glu
50          55          60

Arg Asn Gly Ala Arg Leu Val Pro Gly Tyr Pro Pro Ile Gln Arg Met
65          70          75          80

Val Leu Pro Thr Ile Ala Leu Pro Arg His Phe Ile Asp Lys Val Val
85          90          95

Val Glu Glu Ala Met Asn Gly Tyr Asn Val Arg Leu Val Met Phe His
100         105         110

Arg Lys Leu Leu Ala Val Thr Arg Gly Gly Phe Ile Cys Pro Tyr Thr
115         120         125

Thr Ala Arg Leu Glu Arg Leu Ile Gly Gly Arg Val Arg Glu Leu Phe
130         135         140

Arg Glu Ile Asp Pro Glu Thr Tyr Thr Ile Ala Gly Glu Val Val Gly
145         150         155         160

Leu Glu Asn Pro Tyr Thr Arg Tyr Phe Tyr Pro Glu Ala Pro Arg Phe
165         170         175

Asp Tyr Phe Val Phe Asp Leu Phe His Glu Leu Lys Pro Leu Pro Pro
180         185         190

Leu Glu Arg Asn Glu Leu Leu Glu Lys Tyr Gly Leu Lys His Val Arg
195         200         205

Leu Leu Gly Val Ile Asp Lys Asn Asp Val Glu Met Phe Lys Gln Ile
210         215         220

Val Ala Glu Leu Asp Arg Glu Gly Arg Glu Gly Val Val Ala Lys Asp
225         230         235         240

Pro Glu Tyr Arg Val Pro Pro Leu Lys Tyr Thr Thr Ser Ala Val Asn
245         250         255

Ile Gly Asp Val Arg Tyr Gly Met Arg Phe Phe Met Glu Glu Gly Arg
260         265         270

Ser Phe Leu Phe Ser Arg Leu Leu Arg Glu Leu Phe Arg Ala Tyr Glu
275         280         285

Glu Gly Phe Gly Asp Ala Gln Leu Glu Lys Leu Ala Leu Glu Phe Gly
290         295         300

Arg Ala Ala Thr Glu Pro Ala Leu Glu Ser Ile Arg Lys Val Ala Met
305         310         315         320

Gly Asp Met Leu Tyr Glu Glu Phe Glu Leu Val Phe Ala Asp Glu Val
325         330         335

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<210> SEQ ID NO 7
<211> LENGTH: 390
<212> TYPE: PRT
<213> ORGANISM: artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: four-site mutant of Hyperligase
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (79)..(79)
<223> OTHER INFORMATION: Xaa can be any amino acid except Arg
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (281)..(281)
<223> OTHER INFORMATION: Xaa can be any amino acid except Arg
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (370)..(370)
<223> OTHER INFORMATION: Xaa can be any amino acid except Lys
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (372)..(372)
<223> OTHER INFORMATION: Xaa can be any amino acid except Lys
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<400> SEQUENCE: 7

Met 1	Thr	Trp	Ile	Lys 5	Asn	Pro	Glu	Pro	Trp 10	Met	Val	Asn	Leu	Val 15	Ala
Glu	Lys	Leu	Gly 20	Leu	Asp	Val	Glu	Arg 25	Val	Glu	Thr	Leu	Ala 30	Arg	His
Gly	Thr	Ile 35	Arg	Phe	Arg	Gly	Tyr 40	Arg	Asp	Val	Val	Tyr 45	Ala	Leu	Leu
Arg	Arg 50	Glu	Ile	Ala	Gly	His 55	Pro	Glu	Gly	Thr	Val 60	Val	Leu	Leu	Glu
Arg 65	Asn	Gly	Ala	Arg	Leu 70	Val	Pro	Gly	Tyr	Pro 75	Pro	Ile	Gln	Xaa	Met 80
Val	Leu	Pro	Thr	Ile 85	Ala	Leu	Pro	Arg	His 90	Phe	Ile	Asp	Lys	Val 95	Val
Val	Glu	Glu	Ala 100	Met	Asn	Gly	Tyr	Asn 105	Val	Arg	Leu	Val	Met	Phe	His
Arg	Lys 115	Leu	Leu	Ala	Val	Thr	Arg	Gly 120	Gly	Phe	Ile	Cys 125	Pro	Tyr	Thr
Thr 130	Ala	Arg	Leu	Glu	Arg	Leu 135	Ile	Gly	Gly	Arg	Val 140	Arg	Glu	Leu	Phe
Arg 145	Glu	Ile	Asp	Pro	Glu 150	Thr	Tyr	Thr	Ile	Ala 155	Gly	Glu	Val	Val	Gly 160
Leu	Glu	Asn	Pro	Tyr 165	Thr	Arg	Tyr	Phe	Tyr 170	Pro	Glu	Ala	Pro	Arg	Phe
Asp	Tyr	Phe 180	Val	Phe	Asp	Leu	Phe	His 185	Glu	Leu	Lys	Pro	Leu	Pro	Pro
Leu	Glu	Arg 195	Asn	Glu	Leu	Leu	Glu	Lys 200	Tyr	Gly	Leu	Lys 205	His	Val	Arg

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Leu	Leu	Gly	Val	Ile	Asp	Lys	Asn	Asp	Val	Glu	Met	Phe	Lys	Gln	Ile
210						215					220				
Val	Ala	Glu	Leu	Asp	Arg	Glu	Gly	Arg	Glu	Gly	Val	Val	Ala	Lys	Asp
225					230					235					240
Pro	Glu	Tyr	Arg	Val	Pro	Pro	Leu	Lys	Tyr	Thr	Thr	Ser	Ala	Val	Asn
				245					250					255	
Ile	Gly	Asp	Val	Arg	Tyr	Gly	Met	Arg	Phe	Phe	Met	Glu	Glu	Gly	Arg
		260						265					270		
Ser	Phe	Leu	Phe	Ser	Arg	Leu	Leu	Xaa	Glu	Leu	Phe	Arg	Ala	Tyr	Glu
		275					280					285			
Glu	Gly	Phe	Gly	Asp	Ala	Gln	Leu	Glu	Lys	Leu	Ala	Leu	Glu	Phe	Gly
	290					295					300				
Arg	Ala	Ala	Thr	Glu	Pro	Ala	Leu	Glu	Ser	Ile	Arg	Lys	Val	Ala	Met
305					310					315					320
Gly	Asp	Met	Leu	Tyr	Glu	Glu	Phe	Glu	Leu	Val	Phe	Ala	Asp	Glu	Val
				325					330					335	
Glu	Leu	Glu	Glu	Phe	Thr	Ser	Tyr	Met	Ala	Glu	Leu	Gly	Val	Asp	Ile
			340					345					350		
Val	Val	Val	Ser	Thr	Ser	Arg	Glu	Asp	Glu	Gly	Leu	Arg	Ala	Arg	Met
		355					360					365			
Arg	Xaa	Ile	Xaa	Asp	Thr	Trp	Ile	Gln	Leu	Arg	Lys	Val	Leu	Asp	Thr
	370					375					380				
Gly	Leu	Ser	Pro	Val	Asp										
385					390										

1. A nucleic acid ligase, which comprises an amino acid sequence having a mutation at one or more positions selected from positions 79, 281, 370 and 372 compared to the amino acid sequence of SEQ ID NO: 1.

2. The nucleic acid ligase of claim 1, wherein the mutation at position 79 is the substitution of Arg with Ala, the mutation at position 281 is the substitution of Arg with Ala, the mutation at position 370 is the substitution of Lys with any natural amino acid except Lys, and the mutation at position 372 is the substitution of Lys with any natural amino acid except Lys.

3. The nucleic acid ligase of claim 1, which comprises an amino acid sequence selected from SEQ ID NOs: 2-7.

4. The nucleic acid ligase of claim 3, which comprises the amino acid sequence of SEQ ID NO: 2 or 3, wherein Xaa is a non-polar amino acid, preferably Ala; or, which comprises the amino acid sequence of SEQ ID NO: 4 or 5.

5. The nucleic acid ligase of claim 3, comprising the amino acid sequence of SEQ ID NO: 6.

6. The nucleic acid ligase of claim 5, wherein Xaa at positions 370 and 372 is Ala.

7. The nucleic acid ligase of claim 3, which comprises the amino acid sequence of SEQ ID NO: 7, wherein Xaa at position 79 or 281 is a non-polar amino acid, preferably Ala.

8. The nucleic acid ligase of claim 7, wherein Xaa at positions 370 and 372 is a non-polar amino acid, preferably Ala.

9. A nucleic acid molecule encoding the nucleic acid ligase of claim 1.

10. A vector comprising the nucleic acid molecule of claim 9.

11. A recombinant cell into which the nucleic acid of claim 9 has been introduced.

12. A composition for ligating single-stranded DNAs and/or RNAs, comprising the nucleic acid ligase of claim 1.

13. A kit for ligating single-stranded DNAs and/or RNAs, comprising the nucleic acid ligase of claim 1.

14. A method for producing ligated DNAs and/or RNAs, comprising using the nucleic acid ligase of claim 1 to ligate single-stranded DNAs and/or RNAs.

15. A recombinant cell into which the vector of claim 10 has been introduced.

* * * * *